

Secondary 1 (Grade 7) – GEP Practice

2019 & 2020

Contest Problems with Full Solutions

Authors:

Henry Ong, BSc, MBA, CMA
Merlan Nagidulin, BSc

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About the Authors:

Henry Ong is the Founder of SASMO and President of Singapore International Mastery Contest Centre (SIMCC). He conducts Math Olympiad classes, not only in Singapore but also around the 20 countries that have joined SASMO. He now travels round the world giving out awards and scholarships to students and teachers. He is now promoting new contests, International Scholastic Trust Teachers' Institute (ISTTI) and International Junior Honor Society (IJHS).

In February 2020, Henry was appointed Co-Chair of the Council Administration of Cambodian Mathematical Society (CA_CMS). Henry will share and use his great wealth of experience, expertise, and international networks to contribute effectively to: (1) promoting STEM: Science, Technology, Engineering, and Mathematics; (2) organizing and undertaking multilateral partnership programs; and (3) strengthening and improving the public-private partnership for development of STEM, research and development, and research and innovation as well as teacher professional development in Cambodia with relevant stakeholders in close collaboration and consultation with relevant leadership and management of Ministry of Education, Youth and Sport, especially Cambodian Mathematical Society (CMS), New Generation Pedagogical Research Center (NGPRC), National Institute of Education(NIE) and Teacher Training Department (TTD).

Merlan Nagidulin is the Academic Director for Math and Science Olympiad and General Manager of SIMCC. Merlan graduated with a Bachelor of Science, Honours in Mathematical Sciences from Nanyang Technological University. Merlan won 2 gold medals in Kazakhstan National Math Olympiad and was awarded a bronze medal in the most prestigious International Math Olympiad (IMO Slovenia 2006).

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Noble Education Pte Ltd – 55 Lor L Telok Kurau, 03-69, Bright Centre, Singapore 425500
SIMCC – 199A Thomson Road, Goldhill Centre, Singapore 307636 Website: www.simcc.org

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International Scholastic Trust President's Message

Dear SIMCC contestants, parents, teachers, and partners,

I hope all of you are safe and healthy. We have all gone through so much within the past year, and we should rejoice - we are all alive and well. Despite all the groom and predictions of the worst recession to hit us very soon, we at SIMCC, are expanding globally – adding another 14 country partners to a total of 35 countries and territories within the last 10 months, and setting up sales offices in Cambodia and India, in addition to taking over the ICAS competitions and staff from the University of New South Wales Global Office based in Hong Kong and Macau.

Two years, Sophie Koh, our Senior Director of IT, Events and Operations left Singapore Polytechnic as a Senior Lecturer of IT to join us. She has almost digitally transformed most of our operations and contest systems. This will provide valuable big data in formative and summative assessments for teachers to measure their students and inform their teaching. Parents and students will also receive Detailed Performance Reports (DPR) in our Member Development Portal (MDP) to improve their Academic Skills. Teachers and their schools will be able to download Students' Analytical Performance Reports (SAPR). SIMCC together with our partners will be conducting more Webinars to prepare teachers and parents to use our contests and data to arm their wards with Skills and Recognition for SUCCESS.

From February 16 to April 30, 2021, we are offering the 1st Singapore Math Global Assessment for grades 1 to 11 students. We want to open this up to every country, especially since Singapore Math textbooks are used in schools from over 70 countries. We want to share our rich Singapore Math heritage that has propelled Singapore students to the top of PISA rankings since the 1990s. We also have excellent teachers, Learning Management System (LMS), Math textbooks, assessment books, as well as teacher training to help improve math education worldwide. Reach out to us for help, if you want your students to become Mathematics Heroes!

The global winners of SINGA Math Assessments are invited for the 1st SINGA Math Global Finals, in their home countries on September 11, 2021. The prize for the Overall Champion of grades 1 to 10/11 is a S$600 competition package to compete in the STEAM AHEAD International Junior Mathematics Olympiad (IJMO) to be held in Bali, Indonesia from December 3 to 7, 2021. The prize for Overall Runner-up is S$250 and 2nd Runner-up is S$150, offering a total prize award of S$10,000. Winners of Gold (10%) and Silver (15%) are invited to compete in STEAM AHEAD IJMO 2021.

From 2021, we will be offering most of our contests to grade 12 students – SASMO, DrCT, SIMOC, and STEAM AHEAD. This will pave the way for Grades 12 SIMCC students to apply for SIMCC and STS's US$270,000 endowed 2 scholarships for 4-year undergraduate studies at Southern Illinois University (SIU), Carbondale, Illinois, USA in STEM related fields. SIU has also given SIMCC over 30 SIU International Student Tuition Grants to award to top SIMCC students to pursue their university education. SIMCC will setup our own network for SIMCC scholars at SIU. This will help us expand SIMCC contests in North America and generate revenue for us to fund more scholarships.

Best Regards,
Henry Ong, President

Competition and Prizes

SASMO is devoted and dedicated to bringing a love for Mathematics to students. Unlike most Math Olympiad Competitions, SASMO caters not only to students in the top 5% but to the top 40% instead. It aims to arouse students' interest and enthusiasm for mathematical problem solving, develop mathematical intuition, reasoning and logical thinking, as well as creative and critical thinking. In addition, this can help improve the students' math grades because they can apply problem-solving strategies learnt during the training to their daily school mathematics.

History:

Created in 2006, SASMO is one of the largest Math Olympiads in the Asian region. We have expanded the competition to provide an International platform for students from Primary 2 to Secondary 4, with differentiated contest papers for every level. SASMO awards medals and certificates to the top 40% of participants.

Calculators are not permitted

When a problem introduces a more advanced concept, all necessary definitions are included.

Awards:

Each participant receives a Certificate of Participation or an award certificate for winners below. Each of the top 8%, 12% and 20% of all participants receives a Gold, Silver or Bronze medal and certificate respectively. Each student who achieves a Perfect Score of 85 points receives a Perfect Score certificate, Gold medal and $100.

Introduction

For Students Taking the Math Olympiad Challenge

Congratulations. You have embarked on a journey of scholarship. Competitions like SASMO open many doors for you. Firstly, you learn new and interesting approaches to problem-solving and new topics. Next, you will meet talented students from other schools as you attend trainings and competitions. You build your endurance to "puzzle" out challenging problems and build your reputation as a problem solver. Finally, you will be exposed to various international competitions and scholarship opportunities. Here in Singapore, you increase your chances of getting into a top school via Direct School Admissions (DSA) and entry into the Gifted Education Programme as you compete regularly in high level competitions.

This book is written for the participants in the Singapore and Asian Schools Math Olympiads (SASMO). It helps students to prepare well for the contest and develop higher-order thinking. All problems are designed to help students develop the ability to think mathematically, rather than to teach more advanced or unusual topics. The fun is in how you can see patterns and ways of solving each problem in non-technical ways even though you have not learnt the topic yet!

In addition to the contest problems, the reader is provided with a list of familiar mathematical terms, as well as a review of some of the topics that are likely to be tested in the Olympiad. The book also contains some solved examples to provide different problem-solving techniques, and to familiarize the participant with different types of Olympiad questions. It is advised that the reader spends appropriate time studying these questions and solutions, as they will assist in tackling actual Olympiad problems.

How to Use This Book: Practice daily for 15 minutes per hour instead of 4 hours of learning once a month. Your mind needs to absorb each new thought, and constant practice allows frequent review of previously learned concepts and skills. Together, you can remember many new problem-solving approaches. Try to spend 10 or 15 minutes daily doing two or three problems. This approach should help you minimize the time needed to develop the ability to think mathematically.

Whether you solve a problem quickly or if you are confused, it is worth studying the solutions in this book, because often they offer unexpected insights that can help you understand the problem more fully. After you have invested time – trying to solve each problem any way you can, reviewing our solutions is very effective. Many of the problems in this book can be solved in more than one way. There is always a single answer, but there can be many paths to that answer. Once you solve a problem, go back and see if you can solve it by another method. Then check our solutions to see if any of them differ from yours.

Enjoy working on these challenges and you will soon be in a different league from your peers who have not taken any international competition. We look forward to inviting you if you are a bronze, silver, gold or perfect score medallist for further training.

Problem Solving Procedure

You may go through several phases when solving a problem such as trying to understand the problem, working on a specific approach (planning and attempting), getting stuck and trying to get unstuck, critically examining solutions or communicating. The work may involve going back and forth between these different phases of problem solving.

In solving any problem, it helps to have a working procedure. You might want to consider this four-step procedure: Understand, Plan, Try It, and Look Back.

Understand

Before you can solve a problem, you must first understand it. Read and re-read the problem carefully to find all the clues and determine what the question is asking you to find. What is the unknown? What is the data? What is the condition?

Plan

Once you understand the question and the clues, it's time to use your previous experience with similar problems to look for strategies and tools to answer the question. Do you know a related problem? Look at the unknown! And try to think of a familiar problem having the same or a similar unknown?

Try It

After deciding on a plan, you should try it and see what answer you come up with. Can you see clearly that the steps are correct?
But can you also prove that the steps are correct?

Are you feeling stuck?

Many different approaches can be tried to get unstuck. One approach is to try working a simpler version of the problem, and use the solution to the problem to get insights that are useful in solving the original problem. In the next chapter, we will show some common solving approaches.

If you are discouraged after a few failed attempts, read this quote from the famous scientist, Thomas Edison. An assistant asked, "Why are you wasting your time and money? We have had failure after failure, almost a thousand of them. Why do you continue to pursue this impossible task?" Edison said, "We haven't had a thousand failures, we've just discovered a thousand ways to not invent the electric bulb."

Look Back

Once you've tried a problem and found an answer, go back to the problem and see if you've really answered the question. Sometimes it's easy to overlook something. If you missed something check your plan and try the problem again.
Can you check the result?
Can you check the argument?
Can you derive the result differently?
Can you see it at a glance?

Problem Solving Strategies

1. Change the representation

Using a wrong representation may make a problem impossible to solve.
Strategies of changing representation include drawing a picture and acting it out.

DRAW A PICTURE: By drawing a picture, and visualizing the problem information using it, you will have a clearer understanding of the problem and it will help you to come up an approach to solve the problem that you might not be able to see otherwise.

ACT IT OUT: We are better at thinking in terms of concrete objects and situations than in terms of abstract concepts. If we can act out the situation described in a word problem, we are able to understand the problem better and we may be able to come up with a solution. To do this, we need to use real materials that are easily available to us. Examples can be pencils, coins and other objects we have in the classroom.

2. Make an Organized List or a Table

ORGANIZED LIST: Making an organized list allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

MAKE A TABLE: Making a table allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

3. Create a Simpler Problem

Sometimes we are not able to solve the problem as it is stated, but we are able to solve a similar problem that is similar in some way. For example, the simpler problem may use simpler numbers. Once we solve one or more simpler problems, we may understand the approach that can be used to solve the problems of similar type and may be able to solve the problem that has been given to us.

4. Use Logical Reasoning

Logical reasoning is useful in mathematics problems in various ways. It can be used to eliminate incorrect choices. It can also sometimes be used to conclude the answer directly.

5. Guess and Check

"Guess and Check" strategy can be used on many problems. If the number of possible answers is small, one can use this strategy to come up with the answer very quickly. In some other cases where the number of possible answers is not small, one may still be able to make intelligent guesses and come up with the answer.

6. Working Backwards

Sometimes, it is easier to start with information at the end of the problem and work backwards to the beginning of the problem than the other way around.

SASMO 2019 Secondary 1 (Grade 7) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

The number 2019 is the smallest positive integer that can be written in 6 ways as the sum of squares of 3 prime numbers. One such way is:

$$x^2 + x^2 + 41^2 = 2019$$

What is the value of x?

A. 19
B. 17
C. 13
D. 23
E. None of the above

Question 2

In physics, the kinetic energy (E) of a particle with mass (m) and momentum (p) is described by the relation:

$$E = \frac{p^2}{2m}$$

What will happen to the kinetic energy of a particle if its mass is reduced to half of the original and its momentum is doubled?

A. Unchanged
B. Double
C. Four times the original
D. Eight times the original
E. None of the above

Question 3

Find the sum of the prime factors of $2^8 - 1$.

A. 25
B. 60
C. 17
D. 9
E. None of the above

Question 4

Consider all the integers from 1 to 100. Which of the following is NOT true about them?

A. The number of even numbers is the same as the number of odd numbers.
B. Writing all the digits, the digit "0" is the least occurring digit.
C. There are more even numbers than prime numbers.
D. The number of multiples of 8 is the same as the number of multiples of 9.
E. The average of the 100 numbers is not an integer.

Question 5

What is the smallest four-digit number that is divisible by 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10?

A. 1260
B. 2520
C. 840
D. 5040
E. None of the above

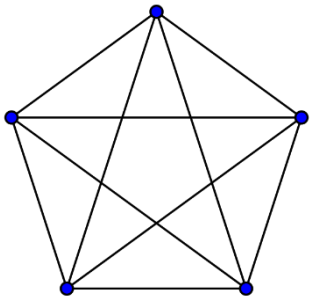
Question 6

The number of Instagram followers of Ariana rose 10% from September to October, then dropped 20% from October to November, then rose 50% from November to December. What was the percentage increase from September to December?

A. 80%
B. 62%
C. 56%
D. 32%
E. None of the above

Question 7

In network science, a "fully connected network" is represented by a graph in such a way that each pair of vertices (called nodes) is connected by exactly one line segment (called a connection). For instance, a fully connected network with 5 nodes is illustrated on the right. This network has 10 connections as shown. How many connections does a fully connected network with 10 nodes have?



A. 45
B. 55
C. 20
D. 66
E. None of the above

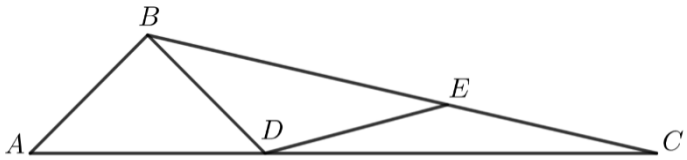
Question 8

The product of 46 integers is 1. Which of the following cannot be the sum of these 46 integers?

A. 38
B. 6
C. 0
D. –14
E. –46

Question 9

In the triangle below, $AB = BD = DE = EC$. Given $\angle ABE = 120°$, what is the measure of $\angle BDA$?



A. $65°$
B. $60°$
C. $45°$
D. $15°$
E. None of the above

Question 10

The operator $\otimes$ acts on two numbers to give the following outcomes:

$$2 \otimes 6 = 44$$

$$4 \otimes 10 = 87$$

$$8 \otimes 18 = 1613$$

$$16 \otimes 22 = 2019$$

What is the value of $32 \otimes 34$?

A. 3266
B. 6440
C. 3225
D. 3233
E. None of the above

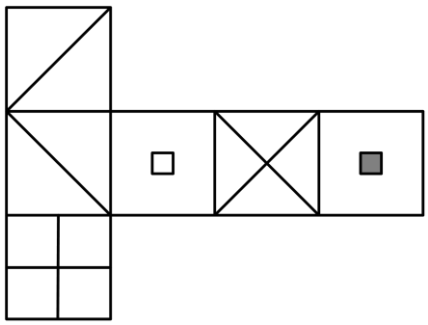
Question 11

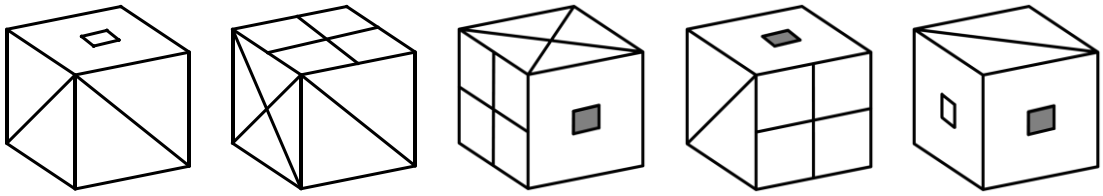
In a camp, each student played at least 2 games. The games were Scavenger Hunt, Animal Charades and Musical Chairs. Twelve students played all three games, 33 played the Scavenger Hunt, 29 played the Animal Charades and 36 played the Musical Chairs. How many students were there in the camp?

A. 62
B. 31
C. 43
D. 110
E. None of the above

Question 12

Which cube below will be formed when the net on the right is folded?





A B C D E

Question 13

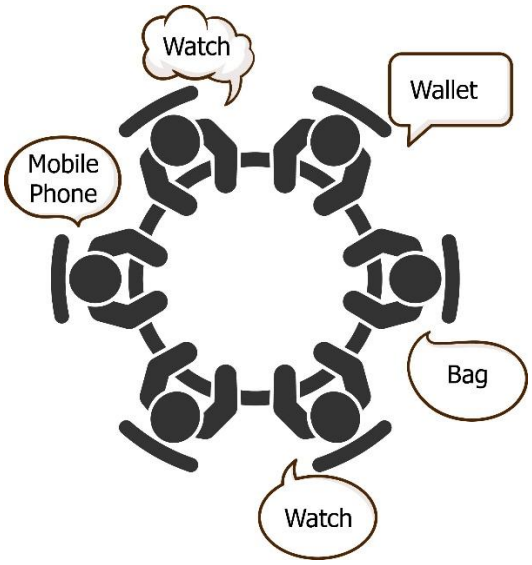
Find the numerator of the following sum in its simplest form.

$$\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\cdots+\frac{511}{512}$$

A. 512
B. 522
C. 3586
D. 4097
E. None of the above

Question 14

A leader asked his 6 followers what is inside the gift box he saw. These 6 followers are sitting around a table. Two of them are truth-tellers and they are sitting next to each other. Two followers are liars and they are sitting next to each other as well. Each of the remaining two followers is either a truth-teller or liar, and they are not sitting next to each other. The responses of the followers are shown in the picture below.



What is inside the gift box?

A. Mobile Phone
B. Watch
C. Wallet
D. Bag
E. Insufficient data

Question 15

Two calendars are the same if all their dates correspond to the same days. The calendar for the year 2019 is the same for the year ____.

(Years 2016, 2012, 2008 and 2004 are the leap years)

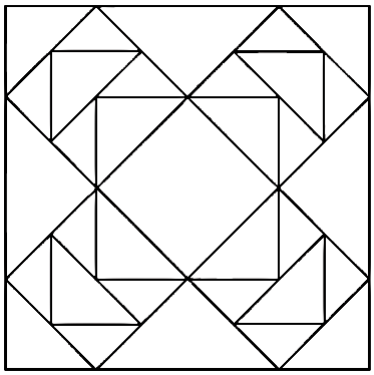
A. 2003
B. 2005
C. 2012
D. 2013
E. 2014

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

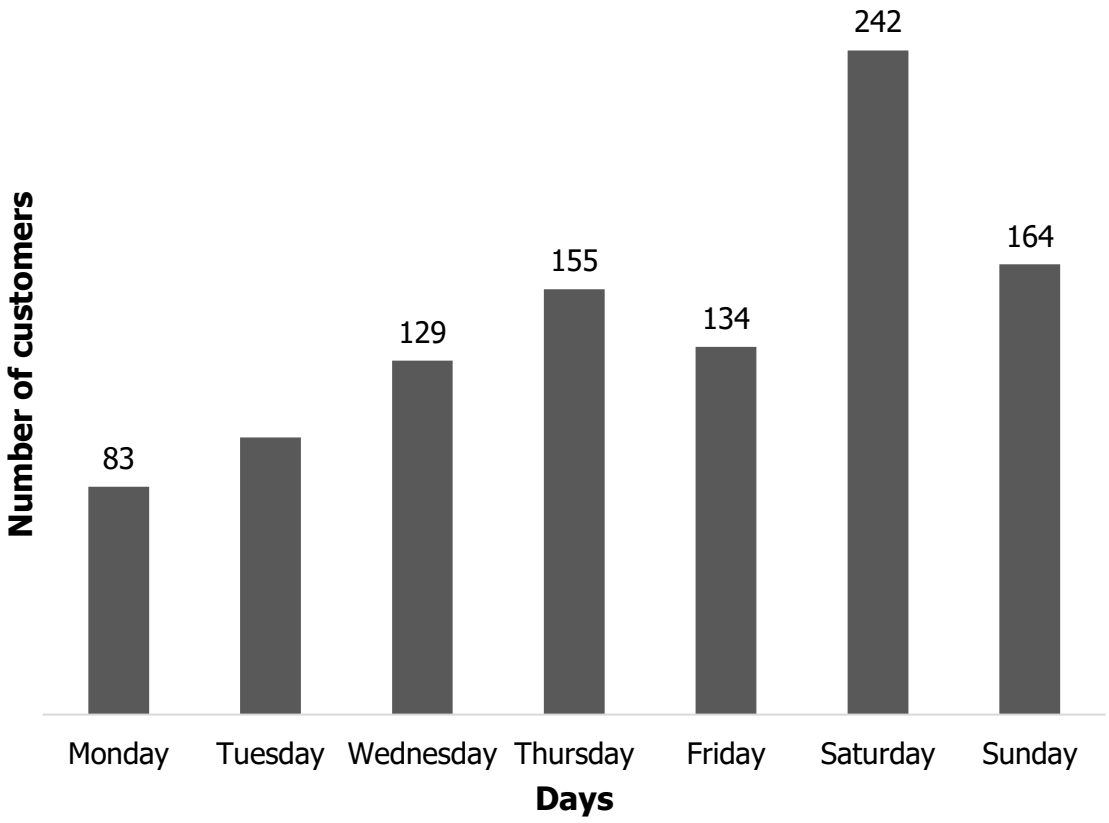
Question 16

How many triangles are there in the figure below?



Question 17

The following bar chart shows the number of customers per day a shop had last week. The average number of customers in the first 4 days was 65% of the average number of customers in the last 3 days. How many customers did the shop have last Tuesday?



Question 18

Annabelle, Betty, Cynthia and Dora were playing a new video game. After the first round of the game:

- Betty and Dora had the same number of gold coins.
- Number of Cynthia's gold coins was half the average number of gold coins of Annabelle and Betty had.
- Cynthia had 10 less gold coins than Dora.
- Annabelle's number of gold coins was half of Betty's.

What was their total number of gold coins after the first round?

Question 19

What is the missing number in the sequence below?

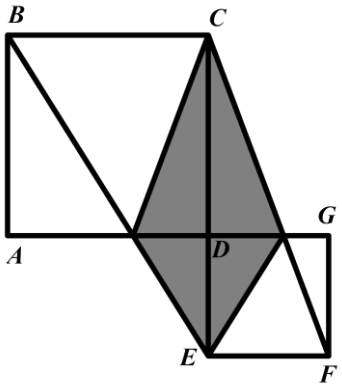
311, 282, 313, 304, 315, 306, 317, ?, 309

Question 20

Find the smallest positive integer n for which $225n$ is a multiple of 4500.

Question 21

In the diagram, $ABCD$ and $DEFG$ are squares with $AB = 15$ cm and $DE = 11$ cm. If $\angle CDG = 90°$, find the area (in cm^2) of the shaded quadrilateral.



Question 22

How many 3-digit numbers have the following properties?

- The number is a multiple of 6.
- It remains a multiple of 6 when its tens and ones digits are switched.

Question 23

Find the sum of the digits of the following product.

$$\underbrace{333\ldots3}_{100\ \ 3\text{s}} \times 2019$$

Question 24

In the following cryptarithm, all the different letters stand for different digits. What is the value of the sum R + E + D + A?

$$\begin{array}{cccccc} & & & R & E & D \\ + & D & E & E & R & \\ \hline & R & D & R & A & \\ \hline \end{array}$$

Question 25

It is given that $159n + 49$ is a four-digit perfect square, where n is a positive integer. Find the value of n.

SASMO 2020 Secondary 1 (Grade 7) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

Find the value of the following.

$$2020 \times 2020 - 2019 \times 2021$$

A. 2020
B. 4040
C. 1
D. 2
E. None of the above

Question 2

Three apples and four oranges cost $10.90, while five apples and seven oranges cost $18.90. How much do nine apples and 13 oranges cost?

A. $20.60

B. $33.10

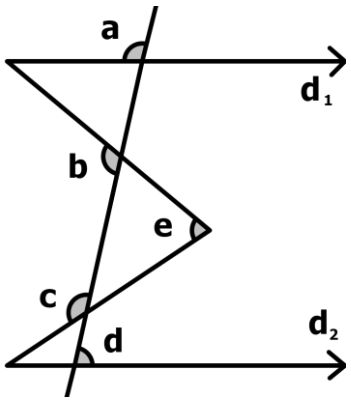
C. $34.20

D. $34.90

E. None of the above

Question 3

In the following diagram, d_1 is parallel to d_2 and $a + b + c + d = 460°$. What is angle e in degrees?



A. $100°$
B. $80°$
C. $70°$
D. $60°$
E. None of the above

Question 4

Find the next term of the following sequence.

1935, 1940, 1948, 1962, 1985, 2020, …

A. 2070
B. 2055
C. 2060
D. 2067
E. None of the above

Question 5

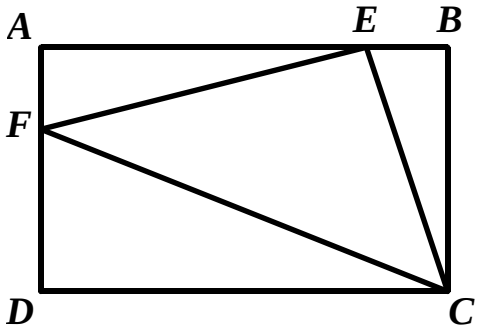
Find the last digit of the following product.

$$2^{2020} \times 3^{2022}$$

A. 6
B. 9
C. 3
D. 4
E. None of the above

Question 6

In the rectangle $ABCD$, point E is on the side AB while point F in on the side AD such that $BE = \frac{1}{4}AE$ and $DF = \frac{2}{3}AD$. What is the ratio of areas of the rectangle $ABCD$ to triangle FEC?



A. 30 : 13
B. 40 : 17
C. 48 : 23
D. 24 : 11
E. None of the above

Question 7

The six-digit number 2X475Y is divisible by 36. How many possible values of X are there?

A. 1
B. 2
C. 3
D. 4
E. None of the above

Question 8

There are roses and lilies at a flower shop. Four roses cost \$7 while three lilies cost \$8. Billy bought some roses and lilies and paid \$86. Which of the following can be the number of roses he bought?

A. 16
B. 24
C. 44
D. 36
E. None of the above

Question 9

How many triples (x, y, z) of positive integers are there such that $x + y + z = 14$?

Take note that (x, y, z), (x, z, y), (y, z, x), (y, x, z), (z, x, y) and (z, y, x) are the same triple.

A. 12
B. 11
C. 13
D. 16
E. None of the above

Question 10

What is the least number of weights required to weigh any objects of integer number of grams from 1 to 35 grams? The weights must be put on one plate while the object is put on the other plate. Also, the weights must be in an integer number of grams.

A. 35
B. 10
C. 7
D. 6
E. None of the above

Question 11

Each face of a soccer ball is either a pentagon or a hexagon. Each pentagonal face is adjacent to five hexagonal faces and each hexagonal face is adjacent to three pentagonal and three hexagonal faces. If the ball has 12 pentagonal faces, how many hexagonal faces are there?

A. 12
B. 20
C. 24
D. 8
E. None of the above

Question 12

Alice, Jane, Philip and Victor were born in 1982, 1983, 1984, 1985 in the cities of Athens, Moscow, Paris and Singapore, though not in that order. It is given that

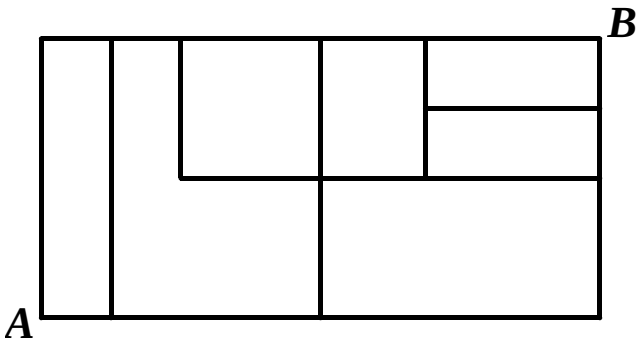
- The person born in Paris is one year older than the one born in Singapore.
- Victor was born one year later than the person born in Paris.
- Philip was born in Moscow.
- Alice was born two years later than Victor was.

Which year was Jane born in?

A. 1982
B. 1983
C. 1984
D. 1985
E. Impossible to determine

Question 13

In the diagram below, an ant can move only upwards or rightwards. How many ways are there for the ant to get from point A to point B along the existing lines?



A. 7
B. 12
C. 8
D. 2
E. None of the above

Question 14

Henry has a bag with 13 yellow, seven brown, 25 red and ten green balls. All the balls are of the same size and shape. What is the least number of balls Henry needs to take out without looking to make sure that he gets three different coloured balls?

A. 3
B. 20
C. 39
D. 48
E. None of the above

Question 15

How many four-digit numbers are there between 3700 and 9600 that can be formed using only the digits 3, 7, 5, 6, 0 or 9 without repetition of any digits?

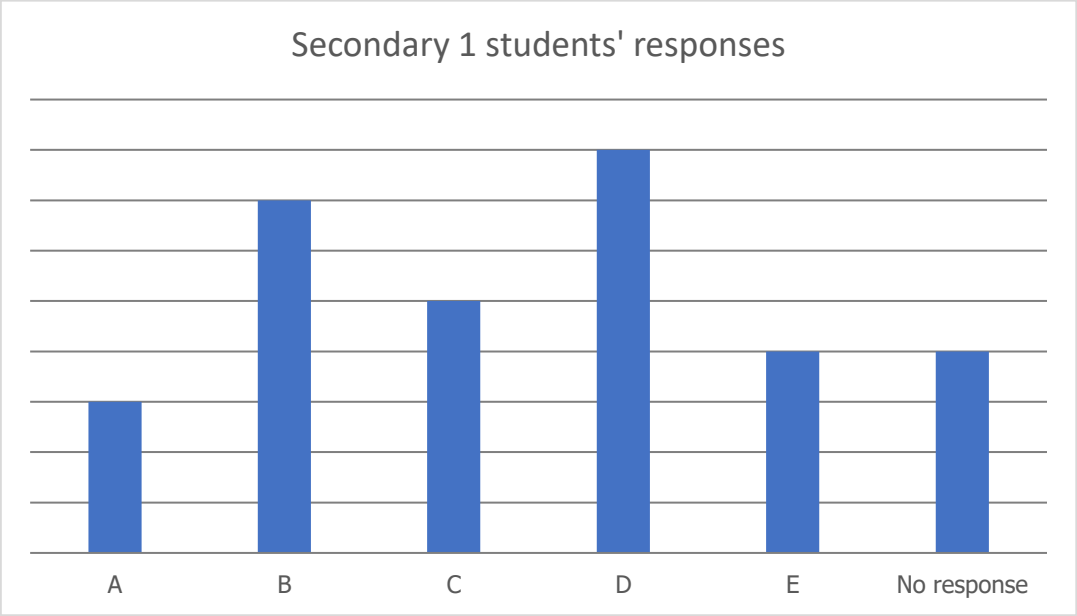
A. 300
B. 100
C. 264
D. 240
E. None of the above

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

Question 16

The following bar chart shows the responses made by all Secondary 1 students from Integrity Secondary School in a multiple-choice question. All the horizontal lines are equally spaced. The correct answer scores 2 points, 0 points for no response and –1 point for the wrong answer. The total points scored by all the students is –120. If the correct answer is B, how many students solved the question correctly?



Question 17

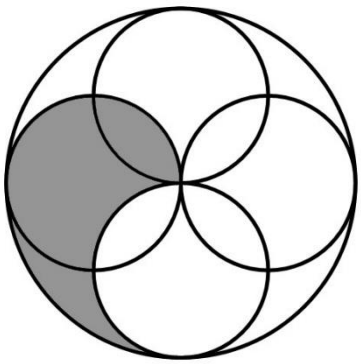
Andrea and Claire left from Woodlands and HarbourFront respectively and travelled towards each other at the same time. When Andrea arrived at HarbourFront, Claire needed to travel for another 5 km more to reach Woodlands. If Andrea travelled 20% faster than Claire, find the distance, in km, between Woodlands and HarbourFront.

Question 18

Find the smallest positive integer n for which $3993n$ is a multiple of 2475.

Question 19

In the diagram, four identical circles touch the large circle and pass through the centre of the large circle. If the diameter of the large circle is 28 cm, find the area (in cm^2) of the shaded region. (Use $\pi = \frac{22}{7}$)



Question 20

Denote $\{n\} = 1 + 3 + 5 + \cdots + (2n - 1)$. For example, $\{5\} = 1 + 3 + 5 + 7 + (2 \times 5 - 1)$. Given that $\frac{\{m\}}{m} + m = 2020$, what is the value of m?

Question 21

What is the smallest positive integer that has exactly 3 odd divisors and 3 even divisors?

Question 22

The operator $\wedge$ acts on two integers to give the following outcomes:

$$4\wedge 8 = 2412$$

$$5\wedge 9 = 2812$$

$$1\wedge 7 = 1618$$

$$2\wedge 3 = 103$$

What is the value of $6\wedge 7$?

Question 23

Peter has five different books of different subjects to be placed on a single-decked shelf. He does not want to place the Physics book next to the Biology one. In how many ways can he place all his books?

Question 24

Peter and Frank together can build a house in 12 days. Frank and George together can build the same house in six days. It is given that each of them works exactly nine hours per day. If Peter and George together can build the house in 6.5 days, how long (in hours) will Peter, Frank and George work together to build the house?

Round your answer to the nearest hour.

Question 25

In the following cryptarithm, all the different letters stand for different digits.

				W	I	N
+	S	A	S	M	O	
	M	E	D	A	L	

If N = 7 and M = 6, find the value of the sum S + A + S + M + O.

END OF PAPER

Solutions to SASMO 2019 Secondary 1 (Grade 7)

Question 1

$x^2 + x^2 + 41^2 = 2019$

$2x^2 + 1681 = 2019$

$2x^2 = 2019 - 1681 = 338$

$x = \sqrt{338 \div 2} = \mathbf{13}$

Answer: **(C)**

Question 2

Since its mass is reduced to half of the original and its momentum is doubled, then

$$m_{new} = \frac{m}{2} \text{ and } p_{new} = 2p.$$

$$E_{new} = \frac{p_{new}^2}{2m_{new}} = \frac{(2p)^2}{2 \times \frac{m}{2}} = \frac{8p^2}{2m} = 8 \times \frac{p^2}{2m} = 8 \times E$$

The new kinetic energy will be **eight times the original**.

Answer: **(D)**

Question 3

Factorise $2^8 - 1$ using the formula $a^2 - b^2 = (a - b)(a + b)$.

$$2^8 - 1 = (2^4 - 1) \times (2^4 + 1) = (2^2 - 1) \times (2^2 + 1) \times (2^4 + 1) = 3 \times 5 \times 17$$

The prime factors of $2^8 - 1$ are 3, 5 and 17.

The answer is $3 + 5 + 17 = \mathbf{25}$.

Answer: **(A)**

Question 4

A. True. There are 50 even and 50 odd numbers.
B. True. The digit "0" appears 11 times whereas other digits appear at least 20 times.
C. True. There are 50 even and 25 prime numbers.
D. **NOT True**. There are 12 multiples of 8 and 11 multiples of 9.
E. True. The average is $\frac{50\times101}{100} = \frac{101}{2}$.

Answer: **(D)**

Question 5

The least common multiple of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 is $8 \times 9 \times 5 \times 7 = 2520$, and it is the smallest four-digit number that is divisible by 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

Answer: **(B)**

Question 6

Let the number of Instagram followers of Ariana in September be x.

Month	Percentage change from previous month	Number of followers
September		x
October	10% increase	$1.1x$
November	20% decrease	$0.8 \times 1.1x = 0.88x$
December	50% increase	$1.5 \times 0.88x = 1.\mathbf{32}x$

The percentage increase from September to December was 32%.

Answer: **(D)**

Question 7

Let $a_1, a_2, \ldots, a_{10}$ be the nodes in a fully connected network with 10 nodes. Count the number of connections in the following way:

The connections that contain a_1: $a_1a_2,\ a_1a_3,\ a_1a_4, \ldots,\ a_1a_{10}$ – 9 connections

The connections that contain a_2: $a_2a_3,\ a_2a_4, \ldots,\ a_2a_{10}$ – 8 connections (the connection a_2a_1 was counted in the connections with a_1)

The connections that contain a_3: $a_3a_4,\ a_3a_5, \ldots,\ a_3a_{10}$ – 7 connections (the connection a_3a_1 and a_3a_2 were counted in the connections with a_1 and a_2)

$\vdots$

The connections that contain a_8: $a_8a_9,\ a_8a_{10}$ – 2 connections

The connections that contain a_9: a_9a_{10} – 1 connection

All the connections that contain a_{10} were counted above.

There are $9 + 8 + 7 + \cdots + 1 = \mathbf{45}$ **connections** in a fully connected network with 10 nodes.

Answer: **(A)**

Question 8

To get the product of 1, all integer factors can only be 1 or -1; and the number of the factor (-1) must be even otherwise the product will be -1.

If the sum of the 46 integer factors is 0, then 23 of them must be 1 and the other 23 must be -1. In this case, the product of the 46 factors will be -1. Thus, **0** cannot be the sum of these 46 integers.

Answer: **(C)**

Question 9

Since $AB = BD = DE = EC$, the triangles ABD, BDE and DEC are isosceles triangles.

Let $\angle BDA = x$, then $\angle BAD = x$ and $\angle ABD = 180 - 2x$

$\angle DBE = 120° - \angle ABD = 120° - (180° - 2x) = 2x - 60° = \angle DEB$

$\angle DEC = 180° - \angle DEB = 180° - (2x - 60°) = 240° - 2x$

$\angle ECD = \angle EDC = \frac{180° - \angle DEC}{2} = \frac{180° - (240° - 2x)}{2} = \frac{2x - 60°}{2} = x - 30°$

In the triangle ABC,

$\angle BAD + \angle ABE + \angle ECD = 180°$ or

$x + 120° + x - 30° = 180°$

$2x + 90° = 180°$

$x = \frac{180° - 90°}{2} = 45°$

$\angle BDA = x = \mathbf{45}°$

Answer: **(C)**

Question 10

The pattern is as follows:

$$a \otimes b = (b-2)\left(\frac{a+b}{2}\right)$$

where we don't multiply the brackets, but write them next to each other.

$$2 \otimes 6 = (6-2)\left(\frac{2+6}{2}\right) = 44$$

$$4 \otimes 10 = (10-2)\left(\frac{4+10}{2}\right) = 87$$

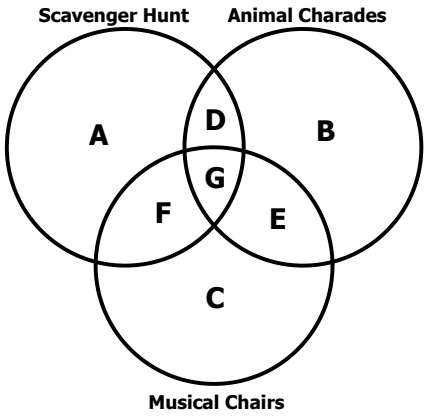
$$8 \otimes 18 = (18-2)\left(\frac{8+18}{2}\right) = 1613$$

$$16 \otimes 22 = (22-2)\left(\frac{16+22}{2}\right) = 2019$$

Therefore, $32 \otimes 34 = (34-2)\left(\frac{32+34}{2}\right) = \mathbf{3233}$

Answer: **(D)**

Question 11



Draw a Venn Diagram as shown in the diagram on the right.

A, B and C are the numbers of students who played only Scavenger Hunt, Animal Charades and Musical Chairs, respectively. Since each student played at least 2 games, then $\mathrm{A} = \mathrm{B} = \mathrm{C} = 0$.

Twelve students played all three games or $\mathrm{G} = 12$.

33 played the Scavenger Hunt or

$\mathrm{A} + \mathrm{D} + \mathrm{G} + \mathrm{F} = 33$ (1)

29 played the Animal Charades or

$\mathrm{B} + \mathrm{D} + \mathrm{G} + \mathrm{E} = 29$ (2)

36 played the Musical Chairs or

$\mathrm{C} + \mathrm{F} + \mathrm{G} + \mathrm{E} = 36$ (3)

Adding equations (1), (2) and (3), we get

$\mathrm{A} + \mathrm{B} + \mathrm{C} + 2\mathrm{D} + 2\mathrm{E} + 2\mathrm{F} + 3\mathrm{G} = 98$ or

$0 + 0 + 0 + 2 \times (\mathrm{D} + \mathrm{E} + \mathrm{F}) + 3 \times 12 = 98$, hence

$\mathrm{D} + \mathrm{E} + \mathrm{F} = (98 - 3 \times 12) \div 2 = 31$

The number of students in the camp is

$$A + B + C + (D + E + F) + G = 0 + 0 + 0 + 31 + 12 = \mathbf{43}.$$

Answer: **(C)**

Question 12

Option A cannot be formed since the faces with diagonals must have different diagonals.
Option B cannot be formed since the 3 shown faces in the option do not share a common vertex when the net is folded.

Option D cannot be formed since the faces with diagonal and middle lines are swapped when the net is folded.

Option E cannot be formed since the faces with white and grey squares must be opposite to each other when the net is folded.

Option C can be formed.

Answer: **(C)**

Question 13

Let

$$S = \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \cdots + \frac{511}{512} \text{ and } B = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{512}$$

$$2B = 2 \times \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{512}\right) = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{256}$$

$$= 1 + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{256}\right) = 1 + \left(B - \frac{1}{512}\right)$$

Hence

$$2B = 1 + \left(B - \frac{1}{512}\right) \text{ and } B = \frac{511}{512}$$

Add S and B

$$S + B = \left(\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \cdots + \frac{511}{512}\right) + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{512}\right)$$

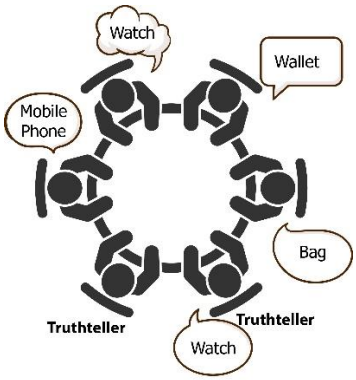
$$= \left(\frac{1}{2} + \frac{1}{2}\right) + \left(\frac{3}{4} + \frac{1}{4}\right) + \cdots + \left(\frac{511}{512} + \frac{1}{512}\right) = 9$$

The numerator of $S = 9 - B = 9 - \frac{511}{512} = \frac{4097}{512}$ is **4097** in its simplest form.

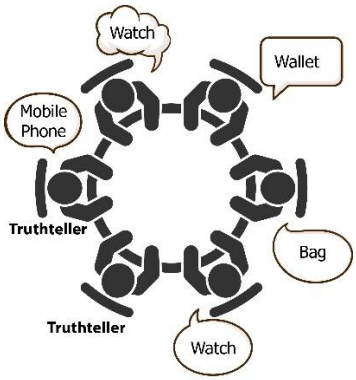
Answer: **(D)**

Question 14

There are only 2 possible ways (shown below) for 2 truth-tellers to sit next to each other.

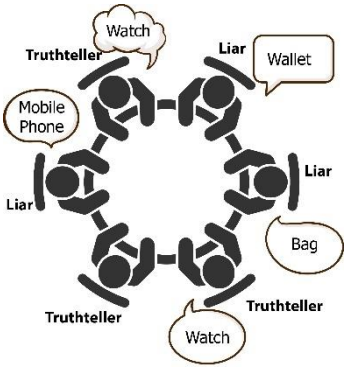


Case 1



Case 2

Let's assume that Case 1 is correct. Then there are 3 truth-tellers and 3 liars as shown on the right. Two liars who responded "Wallet" and "Bag" are sitting next to each other. The remaining two followers, who responded, "Mobile Phone" and "Watch", are also sitting next to each other. This contradicts the question statement. Therefore, Case 1 is incorrect.



From Case 2, we derive that **Mobile Phone** is inside the gift box.

Answer: **(A)**

Question 15

Assume that January 1, 2019 is a Monday. An ordinary year has $365 = (7 \times 52 + 1)$ days and a leap year has $366 = (7 \times 52 + 2)$ days. Thus, each year, January 1st will shift back or forth by 1 or 2 days. When January 1st shifts by 7 or a multiple of 7 days in total, the two calendars will be the same.

From 2019 to 2013, January 1st shifted back by 7 days in total. Hence calendar years 2019 and **2013** are the same.

Year	2019	2018	2017	2016	2015	2014	2013	2012	2011
shift by		–1	–1	–2	–1	–1	–1	–2	–1

Answer: **(D)**

Question 16

The following are the numbers of triangles:

Type of triangle	1-part	9-part
Number	32	4

The total number of triangles is $32 + 4 = \mathbf{36}$.

Answer: **36**

Question 17

Let the number of customers the shop had last Tuesday be x.

The average number of customers in the last 3 days is $\frac{134+242+164}{3} = 180$. The average number of customers in the first 4 days is $0.65 \times 180 = 117$. The average number of customers in the first 4 days is also equal to $\frac{83+x+129+155}{4}$. Hence

$$\frac{83 + x + 129 + 155}{4} = 117$$

Solving the equation, we get $x = \mathbf{101}$.

Answer: **101**

Question 18

Let A, B, C and D be the numbers of coins Annabelle, Betty, Cynthia and Dora had after the first round of the game. Next, we construct an equation from each sentence:

- Betty and Dora had the same number of gold coins. $B = D$
- Number of Cynthia's gold coins was half the average number of gold coins of Annabelle and Betty had. $C = \frac{1}{2} \times \frac{A+B}{2}$
- Cynthia had 10 less gold coins than Dora. $C = D - 10$
- Annabelle's number of gold coins was half of Betty's. $A = \frac{B}{2}$

Solving these equations

$$B = D$$

$$C = \frac{1}{2} \times \frac{A + B}{2}$$

$$C = D - 10$$

$$A = \frac{B}{2},$$

we get $A = 8, B = 16, C = 6 \text{ and } D = 16$.

Their total number of gold coins after the first round was $8 + 16 + 6 + 16 = \mathbf{46}$.

Answer: **46**

Question 19

The pattern is as follows:

31 days in January (the first month of an ordinary year), 28 days in February (the second month of an ordinary year), 31 days in March and so on.

The missing number is **318** which represents the 31 days in August.

Answer: **318**

Question 20

Since $225n$ is a common multiple of 225 and 4500, and n is the smallest positive integer, then $225n$ is the LCM of 225 and 4500.

$225 = 3^2 \times 5^2$

$4500 = 2^2 \times 3^2 \times 5^3$

$\therefore$ LCM of 225 and 4500 $= 2^2 \times 3^2 \times 5^3$

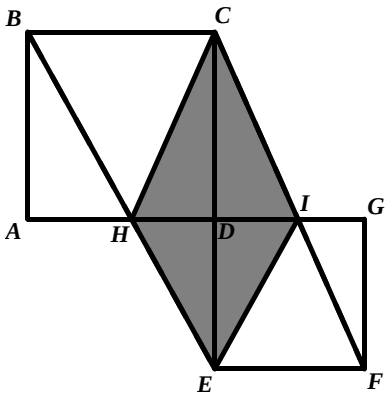
Since $225n$ is the LCM of 225 and 4500, then $225n = 2^2 \times 3^2 \times 5^3$

$$3^2 \times 5^2 \times n = 2^2 \times 3^2 \times 5^3$$
$$n = 2^2 \times 5$$
$$= \mathbf{20}$$

Answer: **20**

Question 21

Let H be the intersection point of AD and BE, and I be the intersection point of CF and GD.



$$Area(ECB) = \frac{CE \times BC}{2} = \frac{(CD + DE) \times BC}{2} = \frac{26 \times 15}{2}$$
$$= 195$$

$$Area(ECF) = \frac{CE \times EF}{2} = \frac{(CD + DE) \times EF}{2} = \frac{26 \times 11}{2}$$
$$= 143$$

$$Area(\ BCH) = \frac{AB \times BC}{2} = \frac{15 \times 15}{2} = \frac{225}{2}$$

$$Area(\ EFI) = \frac{GF \times EF}{2} = \frac{11 \times 11}{2} = \frac{121}{2}$$

$$Area(CHEI) = Area(ECB) + Area(ECF) - Area(\ BCH) - Area(\ EFI)$$
$$= 195 + 143 - \frac{225}{2} - \frac{121}{2} = \mathbf{165}$$

Answer: **165**

Question 22

A multiple of 6 is divisible by 2 and 3. Since a number remains a multiple of 6 when its tens and ones digits are switched, then both tens and ones digits are even. Listing the first eleven 3-digit numbers with the required properties, we have

102, 108, 120, 126, 144, 162, 168, 180, 186, 204, 222,

From here, we can find the pattern which is as follows:

$$102 \overset{+6}{\rightarrow} 108 \overset{+12}{\longrightarrow} 120 \overset{+6}{\rightarrow} 126 \overset{+18}{\longrightarrow} 144 \overset{+18}{\longrightarrow} 162 \overset{+6}{\rightarrow} 168 \overset{+12}{\longrightarrow} 180 \overset{+6}{\rightarrow} 186 \overset{+18}{\longrightarrow} 204 \overset{+18}{\longrightarrow} 222, \ldots$$

6th number: $102 + 6 + 12 + 6 + 18 + 18 = 102 + 60 = 162$

11th number: $102 + 60 \times 2 = 222$

16th number: $102 + 60 \times 3 = 282$

⋮

71st number: $102 + 60 \times 14 = 942$

$$942 \overset{+6}{\rightarrow} \underbrace{948}_{72nd} \overset{+12}{\longrightarrow} \underbrace{960}_{73rd} \overset{+6}{\rightarrow} \underbrace{966}_{74th} \overset{+18}{\longrightarrow} \underbrace{984}_{75th}$$

There are **75** such numbers.

Answer: **75**

Question 23

$$\underbrace{333\ldots3}_{100\ \ 3s} \times 2019 = \underbrace{333\ldots3}_{100\ \ 3s} \times 3 \times 673 = \underbrace{999\ldots9}_{100\ \ 9s} \times 673 = (10^{100} - 1) \times 673$$
$$= 673\,\underbrace{0000\ldots0}_{100\ 0s} - 673 = 672\,\underbrace{999\ldots9}_{97\ \ 9s}327$$

The sum of the digits is $6 + 7 + 2 + 9 \times 97 + 3 + 2 + 7 = \mathbf{900}$.

Answer: **900**

Question 24

From the thousands place, it is obvious that $R = D + 1$.

In the hundreds place addition, the ones digit of $R + E$ is D or the ones digit of $R + E + 1$ is D (if there is a carryover from the tens place addition). Hence, the value of E can only be 8 or 9.

Case 1: $E = 9$

In the tens place addition, there is no carryover of 1 from ones place addition otherwise $R = 9 = E$ which is impossible. Therefore, $R = 8$ $(E + E = 18)$ and $D = R - 1 = 7$ which implies that there is a carryover of 1 from ones place addition. We got the contradiction, so Case 1 is impossible.

Case 2: $E = 8$

In the tens place addition, $R = 6\ or\ 7$ $(E + E = 16\ or\ E + E + 1 = 17)$ and $D = 5\ or\ 6$. Hence, $R + D > 10$ and there is a carryover of 1 from ones place addition. We get $R = 7$ $(E + E + 1 = 17)$, $D = 6$ and $A = 3$.

The sum R + E + D + A is equal to $7 + 8 + 6 + 3 = \mathbf{24}$.

Answer: **24**

Question 25

Let $159n + 49 = a^2$ where $31 < a < 100$

$$159n = a^2 - 49 = a^2 - 7^2 = (a - 7)(a + 7)$$

$$(a - 7)(a + 7) = 159n = 53 \times 3n$$

Hence either $(a - 7)$ or $(a + 7)$ is divisible by 53 which is a prime number. Since $a < 100$, there are only 3 cases:

$(a-7)=53$ or $a=60$

$(a+7)=53$ or $a=46$

$(a+7)=106$ or $a=99$

Only $a=46$ has integer solutions. Thus,

$$n=\frac{a^2-49}{159}=\frac{46^2-49}{159}=\mathbf{13}$$

Answer: **13**

Solutions to SASMO 2020 Secondary 1 (Grade 7)

Question 1

Let $x = 2020$, then $2020 \times 2020 - 2019 \times 2021 = x^2 - (x-1)(x+1) = x^2 - (x^2 - 1) = \mathbf{1}$

Answer: **(C)**

Question 2

Let the cost of each apple be x dollars, and let the cost of each orange be y dollars.

Then, $3x + 4y = 10.90$ (1) and $5x + 7y = 18.90$ (2)

Subtract equation (1) from equation (2) to get $2x + 3y = 8.00$ (3)

Finally, add equation (2) to 2 times of equation (3) to get $9x + 13y = 5x + 7y + 2(2x + 3y) = 18.90 + 8.00 \times 2 = \$\mathbf{34.90}$

Answer: **(D)**

Question 3

Since lines d_1 and d_2 are parallel, $a + d = 180°$ This means $b + c = 460° - 180° = 280°$

Finally, observe that angles $e, (180 - b)°$, and $(180 - c)°$ form a triangle, or

$$e + 180° - b + 180° - c = 180°$$

$$e + 180° = b + c = 280°$$

$$e = \mathbf{100}°$$

Question 4

Find the difference between 2 adjacent terms of the pattern above (e.g. 2nd term – 1st term), then list down the 5 differences to form a new sequence, which is:

$$5, 8, 14, 23, 35$$

Now the pattern is clear, as the differences between 2 adjacent terms in the new sequence are increasing multiples of 3, hence the next term of the new sequence will be $35 + 3 \times 5 = 50$, and the next term of the sequence in question will be $2020 + 50 = \mathbf{2070}$

Answer: **(A)**

Question 5

$$2^{2020} \times 3^{2022} = (2 \times 3)^{2020} \times 3^2 = 6^{2020} \times 3^2$$

Note that the last digit of 6^n will always be 6 if n is a positive integer. Hence, the last digit of $6^{2020} \times 3^2$ is equivalent to the last digit of 6×3^2, which is **4**.

Answer: **(D)**

Question 6

Let AB = DC = $5x$, then AE = $4x$ and BE = x. Let AD = BC = $3y$, then AF = y and FD = $2y$.

$$Area(ABCD) = 15xy$$

$$Area(AEF) = \frac{1}{2} \times 4x \times y = 2xy$$

$$Area(FCD) = \frac{1}{2} \times 2y \times 5x = 5xy$$

$$Area(EBC) = \frac{1}{2} \times x \times 3y = \frac{3xy}{2}$$

And $Area(FEC) = 15xy - 2xy - 5xy - \frac{3xy}{2} = \frac{13xy}{2}$

Hence $Area(ABCD) : Area(FEC) = 15 : \frac{13}{2} = \mathbf{30 : 13}$

Answer: **(A)**

Question 7

The number 2X475Y must be divisible by 9 and 4 for it to be divisible by 36.

Hence the last 2 digits of the number, or 5Y, must be divisible by 4. This means $Y = 2 \; or \; 6$.

If $Y = 2$, the sum of $2 + X + 4 + 7 + 5 + 2$ must be divisible by 9, and $X = 7$ is the only possible choice. If $Y = 6$, the sum of $2 + X + 4 + 7 + 5 + 6$ must be divisible by 9, and $X = 3$ is the only possible choice.

Hence, there are **2** possible values of X.

Answer: **(B)**

Question 8

To get \$86, the number of roses must be divisible by 4 and the number of lilies must be divisible by 3. Let the number of roses be $4 \times a$ and the number of lilies be $3 \times b$. Hence

$$7 \times a + 8 \times b = 86$$

When $a = 1$, b is not a whole number.

When $a = 2$, b is 9.

Checking other values of a from 3 to 12, the possible value of a is 10. Then $b = 9$ and 2 or $3 \times b = 27$ and 6. Thus, none of the options A to D are possible and the answer is **option E**.

Answer: **(E)**

Question 9

List down the triples by the largest number among (x, y, z).

For example, if the largest number is 12, then there is only one possible triple (12, 1, 1).

Largest number	Possible triples	Largest number	Possible triples
12	12, 1, 1	8	8, 3, 3
11	11, 2, 1	7	7, 6, 1
10	10, 3, 1		7, 5, 2
	10, 2, 2		7, 4, 3
9	9, 4, 1	6	6, 6, 2
	9, 3, 2		6, 5, 3
8	8, 5, 1		6, 4, 4
	8, 4, 2	5	5, 5, 4

There are **16** such triples.

Answer: **(D)**

Question 10

To weigh 1-gram objects, we need 1-gram weights.

We can weigh any objects of integer number of grams from 1 to 35 grams using thirty-five 1-gram weights. However, that is not the least number of weights.

Let us optimise to get the least number of weights.

Substitute 18 of 35 one-gram weights to one 18 g weight. So now we have one 18-gram weight and 17 one-gram weights. We still can weigh any weight from 1 to 35 grams.

Next, substitute 9 of 17 one-gram weights to one 9-gram weight. Continue the same process until we have 18 g, 9 g, 4 g, 2 g, 1 g and 1 g weights. We can weigh any weight from 1 to 35 grams using these **6 weights**.

Using mathematics Induction, it can be proved that $[\log_2 N] + 1$ is the least number of weights required to measure N objects with respective weights from 1 to N, where the weights are $\left\lceil \frac{N}{2} \right\rceil, \left\lceil \frac{N}{4} \right\rceil, \left\lceil \frac{N}{8} \right\rceil, \ldots, 1$ and 1.

Answer: **(D)**

Question 11

Each pentagon has 5 sides and each hexagon has 6 sides.

Total number of sides of the pentagons = $5 \times 12 = 60$

Total number of hexagons= $\frac{60}{3} = \mathbf{20}$, since each hexagonal face is adjacent to 3 hexagonal faces.

Answer: **(B)**

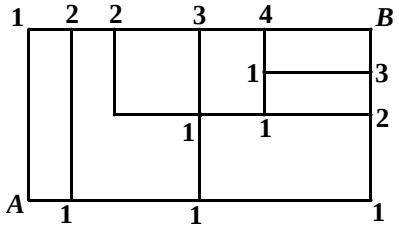
Question 12

Consider the 4 statements above. From statements 1, 2 and 4, it is clear that Victor is the 2nd oldest (born in 1983), and Alice is the youngest (born in 1985). Furthermore, Victor is born in Singapore, Philip is born in Moscow, and Alice cannot be born in Paris as she is the youngest (while the person born in Paris is the oldest). This leaves Jane to be born in Paris. Since she is the oldest, then she must be born in **1982**.

Answer: **(A)**

Question 13

There is only 1 way from point A to the top-left point. Thus, we label that point with '1'. Next, there are 1+1=2 ways to the second left point on top and we label it with '2'. We label the remaining points on the figure similarly as shown below.



From the figure, the number of ways from A to B is 4 + 3 = **7.**

Answer: **(A)**

Question 14

For such questions, it is always best to consider the "worst-case" scenario to determine the least number of balls which needs to be taken out.

Henry takes out 25 balls, unfortunately all 25 balls turn out to be red. Next, Henry takes out 13 more balls, but all of them turn out to be yellow. But after taking out 38 balls, Henry cannot be so "unlucky" anymore, hence the **39th** ball taken out must be either brown or green, making it 3 different coloured balls.

Answer: **(C)**

Question 15

We consider 3 cases.

Case 1: The thousands digit is 3. Then there are 2 choices (7 or 9) for the hundreds digit, 4 choices for the tens digit (0,5,6 or 7/9), depending on the choice of the hundreds digit, and 3 choices for the ones digit, since one of the 4 numbers above will be chosen for the tens digit and digits cannot be repeated.

Total number of choices = $2 \times 4 \times 3 = 24$

Case 2: The thousands digit is 5,6 or 7 (3 choices). Then there are 5 choices (out of the numbers 3,7,5,6,0,9 but one of them will be used for the thousands place) for the hundreds digit, 4 choices for the tens digit, and 3 choices for the ones digit.

Total number of choices = $3 \times 5 \times 4 \times 3 = 180$

Case 3: The thousands digit is 9. Then there are 3 choices (0,3,5) for the hundreds digit, 4 choices for the tens digit (with a similar reasoning to cases 1 and 2), and 3 choices for the ones digit.

Total number of choices = $3 \times 4 \times 3 = 36$

Summing up the 3 cases, we get a total of **240** numbers which fulfil the question's criteria.

Answer: **(D)**

Question 16

Let each space of the bar graph be x students.

Number of students who answered the question correctly = number of students who chose B = $7x$ Hence, a total of $14x$ points were scored from them.

Number of students who answered the question wrongly = number of students who chose A, C, D or E = $3x + 5x + 8x + 4x = 20x$. A total of $-20x$ points were scored from them.

Total number of points = $14x - 20x = -6x = -120$, or $x = 20$.

Number of students who answered the question correctly = $\mathbf{7x = 140}$

Answer: **140**

Question 17

Let the speed of Andrea be $6x$ km/h and speed of Claire be $5x$ km/h

Time taken for Andrea to reach Harbourfront = time taken for Claire to reach 5 km away from Woodlands = $\frac{D}{6x}$, D km is the distance between Woodlands and Harbourfront.

Distance Claire travelled = speed of Claire × time, or $D - 5 = \frac{D}{6x} \times 5x$ which gives $D - 5 = \frac{5}{6}D$

$$D = \mathbf{30}$$

Answer: **30**

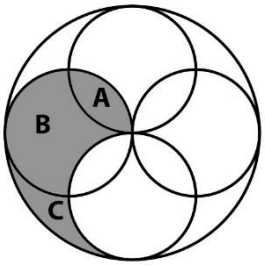
Question 18

First, we factorise 3993 and 2475 to get $3993 = 3 \times 11^3$ and $2475 = 5^2 \times 3^2 \times 11$.

It is given that $\frac{3993n}{2475} = \frac{3 \times 11^3 \times n}{5^2 \times 3^2 \times 11} = \frac{11^2 \times n}{5^2 \times 3} = \frac{11^2 \times n}{75}$ is a positive integer. Hence n is divisible by 75 and the smallest possible value of n is 75.

Answer: **75**

Question 19



Denote three areas A, B and C in the shaded region as shown.

We can notice that the area of the large circle consists of 4 parts of each A, B and C. Hence

$$4A + 4B + 4C = \pi r^2 = \frac{22}{7} \times \left(\frac{28}{2}\right)^2 = 616$$

and the shaded regions is

$$A + B + C = 616 \div 4 = \mathbf{154}.$$

Answer: **154**

Question 20

We attempt to correlate $\{n\}$ with n by finding a pattern.

Observe $\{1\} = 1 = 1^2,\ \{2\} = 1 + 3 = 4 = 2^2,\ \{3\} = 1 + 3 + 5 = 9 = 3^2,\ \{4\} = 1 + 3 + 5 + (2 \times 4 - 1) = 16 = 4^2$

Hence, $\{m\} = m^2$ Now, $\frac{\{m\}}{m} + m = m + m\ (m \text{ is not } 0) = 2m = 2020$

$m = \mathbf{1010}$

Answer: **1010**

Question 21

Let this integer be n.

Since an even number divides n, n must be even. We can then assume that the remaining 2 even divisors of n (besides n itself) are 2 and $2m$, where m is an odd prime.

The smallest possible value of m is 3, so when expressed as a product of 2 numbers, $n = 1 \times n, 2 \times k$ OR $2 \times 2m = 4m$, and $m \times k$ OR $m \times 2m = 2m^2$. k is an arbitrary positive integer, and another ODD divisor of n.

Clearly, $n = 4m = 12$ does not satisfy the question's criteria. On the other hand, $n = 2m^2 = \mathbf{18}$ has 3 odd factors and 3 even factors.

Answer: **18**

Question 22

The pattern is as follows: $a\Lambda b = [2(a+b)][3(b-a)]$

where we don't multiply the square brackets, but write these numbers next to each other. E.g. $4\Lambda 8 = [2(8+4)][3(8-4)] = 2412$

This is a very common "trick" in SASMO, and must be known by heart!

$6\Lambda 7 = [2(6+7)][3(7-6)] = \mathbf{263}$

Answer: **263**

Question 23

Total number of ways to place the 5 books without restraint $= 5! = 120$

We now want to find the number of ways where the Physics and Biology books must be placed together. Treat the adjacent physics and biology books as 1 "multi-book", and this multi-book can be arranged in 2 different ways (by swapping the positions of the Physics and Biology books).

This "multi-book" can be arranged with the 3 other books in a total of $2 \times 4! = 48$ ways.

Total number of ways not to place the physics book next to the biology book is $120 - 48 =$**72**

Answer: **72**

Question 24

Let Peter, Frank and George be able to build $\frac{1}{p}, \frac{1}{F}$ and $\frac{1}{G}$ of a house in 9 hours, or one work-day. Then

$$\frac{1}{P}+\frac{1}{F}=\frac{1}{12}, \frac{1}{F}+\frac{1}{G}=\frac{1}{6}, \frac{1}{P}+\frac{1}{G}=\frac{1}{6.5}=\frac{2}{13}$$

$\frac{1}{P}+\frac{1}{F}+\frac{1}{G}$ can build $\frac{\frac{1}{12}+\frac{1}{6}+\frac{2}{13}}{2}=\frac{21}{104}$ of a house in 9 hours.

Number of hours these 3 people must work together to build a house= $\frac{9}{(\frac{21}{104})}=\frac{312}{7}=$ **45** (to the nearest hour)

Answer: **45**

Question 25

S = 5 because S≠M and there can be a carrying over of maximum 1 to S, so S + 1 = 6. A = 9 because A must carry a 1 over to S, so A + 1 ≥ 10. When A = 9, E = 0.

Substitute N = 7, M = 6, S = 5, A = 9 and E = 0 to get the following:

			W	I	7
+	5	9	5	6	O
	6	0	D	9	L

We are now left with 5 numbers which W, I, O, D and L represent, namely 1,2,3,4 and 8. With these numbers, there are only 2 cases left to consider for 7 + O = L.

Case 1: O = 1 and L = 8. Then I = 3 for I + 6 = 9. That leaves W + 5 = D + 10 (remember that D needs to carry a 1 over to A) and (W,D) = (2,4) or (4,2). This is impossible.

Case 2: O = 4 and L = 1. Then I = 2 for 1 + I + 6 = 9. Finally, W = 8 and D = 3 for W + 5 = D + 10

S + A + S + M + O = 5 + 9 + 5 + 6 + 4 = **29**

Answer: **29**